

Methods and Determination of Quadrupole β_2 Deformation Parameters Across Experimental Techniques

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Abstract

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1 Introduction and Theory

This paper is a compact discussion about how the quadrupole deformation parameters β_2 are measured from reduced transition probabilities $B(E2)$ for a 0^+ to 2^+ transition. Specifically, the Coulomb excitation methodology is the main driver for this measurement, along with simulation tools such as GOSIA. This chapter provides a brief discussion of the theoretical background for defining β_2 and $B(E2)$, including their relationship and how to measure them. The following theoretical background entails starting with the shell model breakdown and how it leads to cases of nonspherical considerations, what the quadrupole formation is, the intrinsic quadrupole moment, and how this is all related to the transition probability $B(E2)$ and deformation parameters β_2 .

1.1 Quadrupole Deformation Overview

The story of nuclear structure always starts with the shell model. Essentially, the shell model is an attempt to explain nuclear structure (how nucleons fill orbitals), in which protons and neutrons in isotopes can have specific orbital filling schemes (see Fig 1) that explain specific parities, total angular momenta, and other properties such as magnetic and quadrupole moments. Most importantly, the shell model explains how certain isotopes are more stable than others due to closed shells, or what we call "magic numbers". As shown in Figure 1, the shell model is governed by a Hamiltonian combining the harmonic oscillator, Woods-Saxon potential, and spin-orbit coupling as a mean field potential. This is important to note because the combination of all three is what provided correct approximations for achieving magic numbers [1]. A single-particle approximation is not an appropriate way to approach nuclear structure. One of the ways the shell model breaks down—leading to the presence of non-spherical nuclear shapes—is the "island of inversion"; essentially, certain isotopes located near the neutron drip line no longer exhibit expected magic numbers (ex, ^{32}Mg see Fig. 2). This failure signals the emergence of strong correlations and collective behavior, leading us to think of nuclei as deformable many-body systems rather than spherical collections of independent particles.

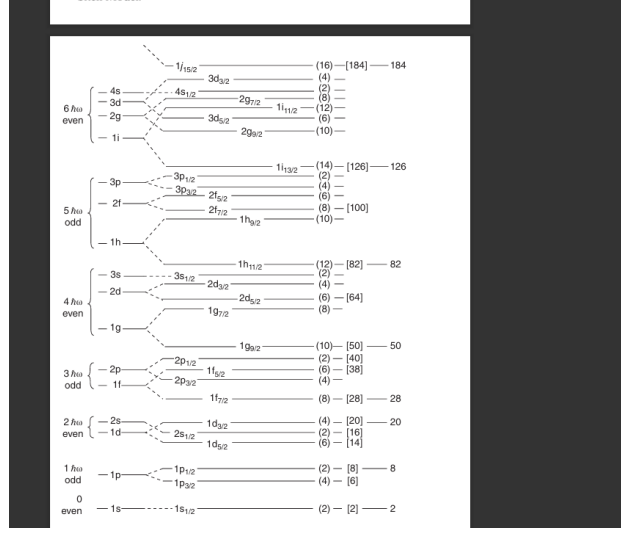


Figure 1: Example of a shell model spectroscopic chart based on how nucleons fill orbitals (harmonic oscillator + Woods-Saxon + spin/orbit). Image taken from [1].

When such deviations from spherical symmetry occur, collective motion becomes the dominant feature of nuclear structure. This includes both rotational and vibrational degrees of freedom. In particular, quadrupole deformation arises when the nuclear shape deviates from spherical symmetry, assuming either a prolate (rugby-ball) or an oblate (pancake-like) configuration.

The shape of the nuclear surface can be mathematically described using spherical harmonics. For quadrupole deformation, the nuclear surface is expanded as:

$$R(\theta) = R_0 (1 + \beta_2 Y_{20}(\theta)) , \quad (1)$$

where:

- R_0 is the radius of a spherical nucleus,
- β_2 is the quadrupole deformation parameter,
- $Y_{20}(\theta) = \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$ is a second-order Legendre polynomial (spherical harmonic).

The deformation parameter β_2 quantifies the degree and nature of the deformation. A positive β_2 implies a prolate shape, while a negative β_2 implies an oblate shape.

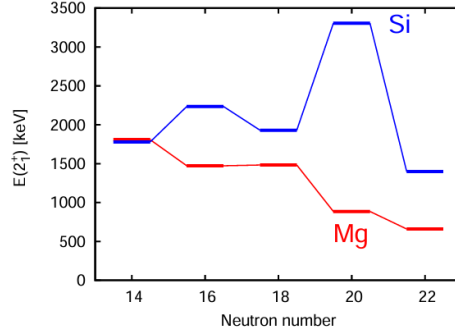


Figure 2: Excitation energies of the first 2^+ state for even-even Si (blue) and Mg (red) isotopes as a function of neutron number. The trend shows enhanced collectivity (lower $E(2^+)$) in neutron-rich Mg isotopes, consistent with quadrupole deformation and breakdown of the shell model near the so-called island of inversion. Image from [2].

1.2 Intrinsic Quadrupole Moment

One of the essential features, or "measurements," that arise from collective motion—or, more specifically, from non-spherical assumptions about nuclear shape—is the intrinsic quadrupole moment, Q_0 . This quantity characterizes the degree to which the nuclear charge distribution deviates from spherical symmetry in the body-fixed (intrinsic) frame of a deformed nucleus.

For axially symmetric deformed nuclei, the intrinsic quadrupole moment is defined as:

$$Q_0 = \int (3z^2 - r^2) \rho(\vec{r}) d^3r \quad (2)$$

Where $\rho(\vec{r})$ is the charge density distribution, r is the radial distance from the origin, and z is the coordinate along the symmetry axis of the nucleus. This expression compares the elongation along the symmetry axis to the average radius. A positive Q_0 corresponds to a prolate shape (rugby ball-like), while a negative Q_0 implies an oblate shape (pancake-like).

The intrinsic quadrupole moment is not directly measurable in the laboratory frame, but it connects to observable quantities like the spectroscopic quadrupole moment Q or the reduced electric quadrupole transition probability $B(E2)$. For a transition from a 0^+ to 2^+ state, the relation to Q_0 is:

$$B(E2; 0^+ \rightarrow 2^+) = \frac{5}{16\pi} Q_0^2 \quad (3)$$

This equation serves as the bridge between observed electromagnetic transitions and the intrinsic nuclear shape.

Later sections will show how measurements of gamma-ray emission intensities can be used to extract $B(E2)$ values, which are then inverted via this equation to determine Q_0 and, ultimately,

the deformation parameter β_2 .

1.3 Reduced Transition Probability $B(E2)$

The reduced transition probability $B(E2)$ quantifies the probability that a nucleus in an excited state will emit an electric quadrupole (E2) gamma-ray photon and transition to a lower energy state. In particular, the transition from a first excited 2^+ state to the ground 0^+ state in even-even nuclei is a standard probe of quadrupole collectivity. A larger $B(E2)$ value typically indicates greater collectivity and deformation.

Mathematically, the $B(E2)$ value is related to the square of the matrix element of the electric quadrupole operator $\hat{Q}_{2\mu}$:

$$B(E2; I_i \rightarrow I_f) = \frac{1}{2I_i + 1} |\langle I_f || \hat{Q}_2 || I_i \rangle|^2 \quad (4)$$

For the special case of $0^+ \rightarrow 2^+$ transitions in axially symmetric nuclei, this reduces to:

$$B(E2; 0^+ \rightarrow 2^+) = \frac{5}{16\pi} Q_0^2 \quad (5)$$

where Q_0 is the intrinsic quadrupole moment of the nucleus, this relationship allows us to extract internal structure information (e.g., deformation) from experimental gamma-ray measurements.

1.4 Key Relations for Extracting β_2

The quadrupole deformation parameter β_2 is essentially a classification of how non-spherical a specific isotope shape could be in reference to either being prolate ($\beta_2 > 0$) or oblate ($\beta_2 < 0$). Coulomb excitation processes measured β_2 by using Eqns. 4 and 6 in conjunction with each other. A more concrete derivation of reduced transition probabilities and quadrupole deformation parameters can be found via these citations [2, 3].

$$|\beta_2| = \frac{4\pi}{3ZeR_0^2} \sqrt{B(E2; 0_{\text{g.s.}}^+ \rightarrow 2_1^+)}. \quad (6)$$

2 Experimental Methods

2.1 Principle of Coulomb Excitation

2.2 Inverse Kinematics

2.3 Target Considerations

2.4 Experimental Observables

2.5 Detector Systems

3 Methodology for Extracting $B(E2)$ and β_2

3.1 Gamma-Ray Identification

3.2 Efficiency Corrections

3.3 Angular Analysis

3.4 Normalization Procedures

3.5 GOSIA Analysis

3.6 Extraction of E2 Matrix Elements

3.7 Calculation of $B(E2)$

3.8 Calculation of β_2

4 Selected Results from Literature

4.1 ^{30}Mg

4.2 ^{32}Mg

4.3 ^{74}Zn

4.4 ^{76}Zn

5 Conclusion

- [2] Oliver Niedermaier. “Coulomb excitation of neutron-rich magnesium isotopes”. PhD thesis. University of Heidelberg, 2005.
- [3] David Verney. “History of the concept of nuclear shape”. In: *European Physical Journal A* 61 (2025), p. 82. DOI: [10.1140/epja/s10050-025-01545-1](https://doi.org/10.1140/epja/s10050-025-01545-1).